

Advanced (Carolina Reaper)

Q1. Solve the inequality $|x| < 3$.

- (A) $(-\infty, -3) \cup (3, \infty)$
 (B) $(-3, 3)$
 (C) $(-\infty, 3] \cup [3, \infty)$
 (D) $[3, \infty)$
 (E) $(-\infty, -3] \cup [3, \infty)$

Q2. Solve the inequality $|2x - 1| \geq 5$.

- (A) $(-\infty, -2] \cup [3, \infty)$
 (B) $(-\infty, 2] \cup [3, \infty)$
 (C) $(-\infty, -2) \cup (3, \infty)$
 (D) $(-2, 3)$
 (E) $(-2, 3)$

Q3. The solution set of $|x + 2| > 4$ is

- (A) $(-\infty, -6) \cup (2, \infty)$
 (B) $(-\infty, -2) \cup (4, \infty)$
 (C) $(-6, 2)$
 (D) $(-2, 4)$
 (E) $(-\infty, -2) \cup (2, \infty)$

Q4. The graph of $|x| \leq 2$ on a number line is

- (A) All points between -2 and 2 inclusive.
 (B) All points between -2 and 2 exclusive.
 (C) All points greater than or equal to -2.
 (D) All points less than or equal to 2.
 (E) All points between 2 and -2 exclusive.

Q5. Solve $|3x - 1| < 2$.

- (A) $(-\frac{1}{3}, 1)$

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- (A) $(-1, 1)$

- (B) $(-\frac{1}{3}, 1)$

 $(-\frac{1}{3}, 1)$ $(-\frac{1}{3}, 1)$

- (B) $(-\frac{1}{3}, 1)$

 $(-\frac{1}{3}, 1)$ **Q6.** Solve $|x - 4| \geq 3$.

- (A) $(-\infty, 1] \cup [7, \infty)$
 (B) $(-\infty, 1) \cup (7, \infty)$
 (C) $(-\infty, 7] \cup [1, \infty)$
 (D) $(-\infty, -3) \cup (7, \infty)$
 (E) $(-\infty, -3) \cup (3, \infty)$

Q7. The absolute value inequality $|x| > 2$ can be written as

- (A) $x > 2$ or $x < -2$
 (B) $x > 2$ and $x < -2$
 (C) $x < 2$ and $x > -2$
 (D) $x \leq 2$ and $x \geq -2$
 (E) $x \geq 2$ and $x \leq -2$

Q8. A company packages coffee in bags that weigh within 0.5 pounds of the ideal weight of 2 pounds. If x represents the weight, in pounds, of a bag of coffee, the inequality that represents the possible weight of the bag of coffee is

- (A) $|x - 2| < 0.5$
 (B) $|x - 2| > 0.5$
 (C) $|x + 2| > 0.5$
 (D) $x > 2.5$
 (E) $|x| > 2$

Open Ended Questions (FRQ)

Q9. The temperature in a certain city is never more than 25°F from the average temperature. If x represents the temperature and the average temperature is 50°F , write an inequality representing the range of possible temperatures in the city and graph it on the number line.**Q10.** A bakery sells a total of 250 loaves of bread per day. The number of whole wheat loaves sold, x , is related to the number of white loaves sold, y , by the inequality $|x - y| \leq 50$. Write the inequality in terms of x when $y = 200$, and solve for x . Show your work and explain your reasoning.

Chef's Tips

- Provide graph paper for question 9.
- Award partial credit where applicable.
- Remind students to show all work for free response questions.